Finding the Morse Potential using Variational Monte Carlo Techniques

Ved Patel

***Abstract*—In this paper, we discuss the process of finding the Morse potential using variational Monte Carlo techniques. The values found for the bond length and the dissociation energy are 1.40 ± 0.01 and 0.137 ± 0.003.**

I. INTRODUCTION

QUANTUM mechanical observables are inherently probabilistic. The distribution of observables depends on the wavefunction of a system. If we want to be able to predict anything, we need the wavefunction. Finding the wavefunction involves solving the Schrödinger Equation, which in most cases, cannot be done analytically. In this report, we will discuss the process of solving the time-independent Schrödinger Equation to find the Morse potential of a Hydrogen molecule using variational Monte Carlo techniques. To find the Morse potential, we will need to find the ground state.

The ground state of a system is the state in which the energy expectation is minimised. The expectation of the Hamiltonian for an unnormalised wavefunction is given by the following integral:

$$\langle H \rangle = \frac{\int_Q \psi^*(q)\hat{H}\psi(q)dQ}{\int_Q \psi^*(q)\psi(q)dQ}, \tag{1}$$

where $q \in Q$ is an element of the configuration space and the denominator is a normalisation factor.

In this paper, we work our way towards the Hydrogen molecule by considering various validation systems in the following order:

- 1D quantum harmonic oscillator ($Q = \mathbb{R}$)
- 3D hydrogen atom ($Q = \mathbb{R}^3$)
- 3D hydrogen molecule ($Q = \mathbb{R}^6$)

A. 1D Quantum Harmonic Oscillator

This problem involves verifying energy eigenstates of a quantum harmonic oscillator with natural frequency ω and mass m_e. We proceed to change the units of our problem by making the following substitution:

$$u = \sqrt{\frac{\omega m_e}{\hbar}}x, \tag{2}$$

and so, the dimensionless Hamiltonian for this system is as follows:

$$\hat{H} = -\frac{1}{2}\frac{d^2}{du^2} + \frac{1}{2}u^2, \tag{3}$$

where the energy is given in units of $\hbar\omega$.

We know that the theoretical energy eigenvalues E_n and eigenstates $\psi_n(x)$ are given by:

$$E_n = n + \frac{1}{2} \tag{4}$$

$$\psi_n(x) = H_n(x)\exp\left(-\frac{x^2}{2}\right), \tag{5}$$

where H_n are the n_{th} order Hermite polynomials.

We verified these eigenstates by numerically evaluating the time-independent Schrödinger Equation for each eigenstate and comparing the energy eigenvalues given to the theoretical eigenvalues.

B. 3D Hydrogen Atom

This problem involves finding the ground state energy of the hydrogen atom. We have a single electron in 3 dimensions. For our Hamiltonian, we used the Born-Oppenheimer approximation of the atom, which assumes a fixed nucleus. In this case, the configuration q corresponds to a 3 dimensional vector $\mathbf{r}$. The dimensionless Hamiltonian is as follows:

$$\hat{H} = \left[-\frac{1}{2}\nabla^2 - \frac{1}{|\mathbf{r}|}\right], \tag{6}$$

where the respective units of energy and distance are as shown below:

$$1\text{Ha (Hartree)} = \frac{m_e e^4}{(4\pi\varepsilon_0)^2\hbar^2} \tag{7}$$

$$1a_0 \text{ (Bohr Radius)} = \frac{4\pi\varepsilon_0\hbar^2}{m_e e^2} \tag{8}$$

We take the following ansatz:

$$\psi(\mathbf{r}) = e^{-\theta|\mathbf{r}|} \tag{9}$$

We want to find the optimal parameter θ which minimises the expectation value of the Hamiltonian. We know analytically that the minimum energy is -0.5 and that the optimal value of θ is 1, so we can compare our numerical values with these theoretical ones.

C. 3D Hydrogen Molecule

This problem involves finding the Morse Potential for the Hydrogen molecule. In order to do this, we must find the ground state energy of the Hydrogen molecule for various atomic separations. The theoretical form of this potential is as follows:

$$V(r) = D(1 - e^{-a(r-r_0)})^2 - D + 2E_{single}, \tag{10}$$

where D, a and r_0 are all constants to be found and E_{single} is the minimised energy of the hydrogen atom, which we know to be -0.5. D is the dissociation energy and r_0 is the bond length.

We now have 2 particles in 3 dimensions, so our configuration space is 6 dimensional. The position vectors $\mathbf{R}_i$ will denote the positions of the nuclei. The dimensionless Hamiltonian for this problem is as follows:

$$\hat{H} = -\frac{1}{2}\sum_{i=1}^{2}\nabla_i^2 + \frac{1}{r_{12}} + \frac{1}{R_{12}} - \sum_{i,j=1}^{2}\frac{1}{|\mathbf{r}_i - \mathbf{R}_j|}, \tag{11}$$

where r_{12} and R_{12} denote $|\mathbf{r}_2 - \mathbf{r}_1|$ and $|\mathbf{R}_2 - \mathbf{R}_1|$ respectively and $\nabla_i^2 = \partial_{x_i}^2 + \partial_{y_i}^2 + \partial_{z_i}^2$ is the Laplacian with respect to the coordinates of particle i. We are working in the same units as we were for the Hydrogen atom. Our ansatz for this problem now has 3 parameters $\theta = (\theta_1, \theta_2, \theta_3)$ and we can write it like this:

$$\psi(\mathbf{r}) = (t_1 + t_2)\exp\left(-\frac{\theta_2}{1 + \theta_3 r_{12}}\right), \tag{12}$$

where

$$t_1 = e^{-\theta_1(|\mathbf{r}_1 - \mathbf{R}_1| + |\mathbf{r}_2 - \mathbf{R}_2|)}, \tag{13}$$

$$t_2 = e^{-\theta_1(|\mathbf{r}_1 - \mathbf{R}_2| + |\mathbf{r}_2 - \mathbf{R}_1|)} \tag{14}$$

II. Methods

A. Monte Carlo Integration

We can evaluate equation 1 by noticing that it is equivalent to formal definition of expectation value:

$$\langle H\rangle = \langle E\rangle = \int_Q \rho(q)E_l(q)dQ \tag{15}$$

where $\rho = \psi^*\psi$ is the normalised PDF (probability density function) and $E_l = \frac{1}{\psi}\hat{H}\psi$ is the local energy of the configuration q. This expectation value can be approximated using the sample mean as follows:

$$\langle H\rangle \approx \frac{1}{N}\sum_{i}^{N} E_l(q_i), \tag{16}$$

where $q_i \sim \rho(q)$

Then the error is given by the standard error, which comes from the following formula:

$$\Delta\langle H\rangle = \frac{\sigma}{\sqrt{N}} = \frac{\sqrt{\sum_i^N(\langle H\rangle - E_l(q_i))^2}}{N} \tag{17}$$

B. Metropolis Sampling

To find our sample mean for the energy, we must first sample configuration points from our probability density function $\rho(q)$. The sampling technique used for this was the Metropolis algorithm. This algorithm was a nice choice because it allows us to sample from an unnormalised PDF. If we were to normalise the PDF manually, this would involve another integral, which would require far more computation, and introduce more systematic error.

The algorithm is a Markov chain Monte Carlo technique, so it recursively generates samples based on the previous sample. The algorithm initialises with some starting sample q_0 and then for every new step, a new potential sample point q' is proposed based on a certain proposal PDF. This sample is then either accepted or rejected randomly with the following probability:

$$p_{acc} = \min\left(1, \frac{\rho(q')}{\rho(q_i)}\right) \tag{18}$$

where p_{acc} is the probability of accepting point q_{i+1}. If the point is accepted then $q_{i+1} = q'$ and if the point is rejected, then $q_{i+1} = q_i$. The points generated from this algorithm are distributed from the

PDF $\rho(q)$. However, these points are not independent from one another. If there are not enough points to fully represent the distribution, the generated points will show a bias around the starting point. To try and combat this, we run multiple Metropolis chains from different starting points and burn points from the start. This reduces the correlation between the points.

For our proposal distribution, we use a standard normal distribution in the appropriate number of dimensions with mean q_i.

The Metropolis algorithm was tested in 1 dimension to see whether the generated distribution is the target PDF as shown in Fig.1

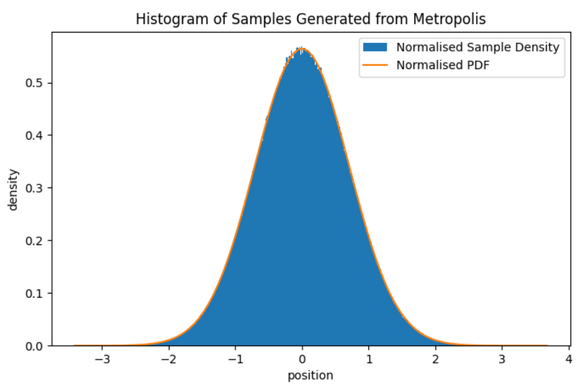

Fig. 1. Histogram of points generated from the Metropolis algorithm. There are 5 metropolis chains each of which generated 10^6 points. The target PDF (before normalisation) is the function e^{-x^2}, which corresponds to the first eigenstate of the quantum harmonic oscillator.

For each Metropolis chain we use burn-in and initialise using a normal distribution of standard deviation 1. These points will span most of the useful information in the PDF, since everything is dimensionless. The first 5 samples are discarded to reduce the effect of the starting point in the Metropolis chain. The effectiveness of the burn in has not been tested. However, there are orders of magnitudes more samples per chain than starting points, so the distribution of starting points will not contribute much to the final distribution.

C. *Finite Difference*

Our expression for the local energy involves the Hamiltonian operator, which contains a Laplacian in one of the terms. We evaluate this numerically using a finite difference scheme. We chose the fourth-order central difference scheme for the 2nd derivative.

This difference scheme is given by the following formula:

$$\frac{\partial^2 \psi}{\partial x^2} \approx \frac{-\psi_2 + 16\psi_1 - 30\psi_0 + 16\psi_{-1} - \psi_{-2}}{12h^2}, \tag{19}$$

where $\psi_i = \psi(\mathbf{r} + ih\hat{\mathbf{x}})$ for some defined step size h. This scheme has a truncation error of order h^2.

As a quick preliminary test, the error against step size for a first derivative finite difference scheme was plotted as shown in Fig.2 This example demonstrates that there exists an optimal step size and difference scheme to minimize error.

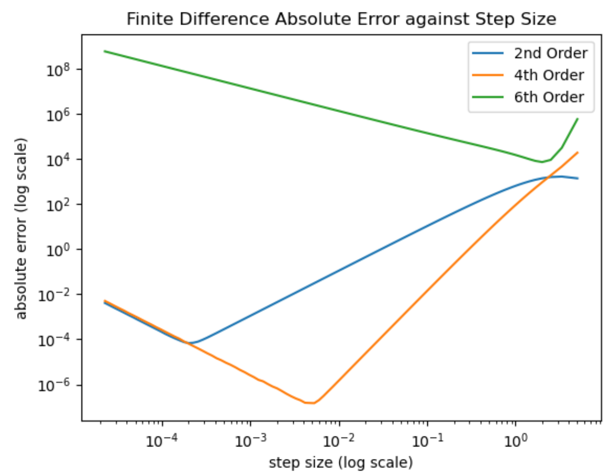

Fig. 2. This is a plot showing the error of different order finite difference approximations of the first derivative. This finite difference was applied on the example function e^x over the range $1 \rightarrow 11$. The error plotted is the mean error across all samples taken for a specific step size.

D. *Gradient Descent*

The method we used to minimise energy expectation was gradient descent. This involves descending a function by taking steps proportional to the gradient as follows:

$$\theta_{n+1} = \theta_n - \alpha \nabla f(\theta)|_{\theta=\theta_n}, \tag{20}$$

for some proportionality factor α. The value of alpha affects the speed of convergence. If α is too small, then each iteration will greatly undershoot the minimum and it will take many iterations to reach the minimum. If α is too big then the steps will overshoot the minimum and the gradient descent will either oscillate around the minimum or it may diverge entirely. Multiple α values were tested to ensure convergence, and a choice was made. The optimal value of α depends greatly on the problem, so the choice of alpha was one that gave consistent convergence for all problems. For the Hydrogen atom and molecule, the choices of α are 0.5 and 0.1, respectively.

Gradient descent makes use of applying the gradient operator on our expected energy $\nabla_\theta \langle H \rangle$. This was calculated analytically using the following formula:

$$\partial_\theta \langle H \rangle = \frac{2}{N} \sum_i^N \left[(E_l(q_i) - \langle H \rangle) - \frac{\partial_\theta \psi}{\psi}|_{q_i} \right] \quad (21)$$

Getting the gradient involves finding the mean gradient of many different samples, so getting the error is a matter of finding the standard deviation of the sample mean gradient in a manner similar to how the error was found in equation (17).

As the gradient decreases, more samples are taken each iteration. This is because when our parameters are far from the minimum, the error in the gradient doesn't matter as much. However, as we get closer to our minimum and our gradient shrinks, we want less error in our gradient to ensure that we don't step away from the minimum due to noise.

This is a similar method to one used in Ref. [2].

III. Results

A. 1D Quantum Harmonic Oscillator

The energy eigenvalues were calculated for the first 5 eigenstates and are shown in Table I. 4 out of 5 of these eigenvalues line up with the theoretical values within 1 standard deviation. The 5th eigenvalue is just over 2 standard deviations away from the theoretical value, which is a reasonable fluctuation considering the possibility of rounding. The samples for each eigenstate were generated from 5 Metropolis chains of length 10^5

Eigenstate Number	Energy Eigenvalue ($\hbar\omega$)
0	$0.500000001 \pm 0.000000001$
1	$1.500000000 \pm 0.000000002$
2	$2.499999995 \pm 0.000000002$
3	$3.499999997 \pm 0.000000002$
4	$4.499999999 \pm 0.000000003$

TABLE I
CALCULATED ENERGY EIGENVALUES FOR DIFFERENT EIGENSTATES

B. 3D Hydrogen Atom

The obtained value of the parameter θ is 1.0000007 ± 0.0000008, which is within 1 standard deviation of the theoretical value. The PDF of the wavefunction of the ground state can be visualised in Fig. 3.

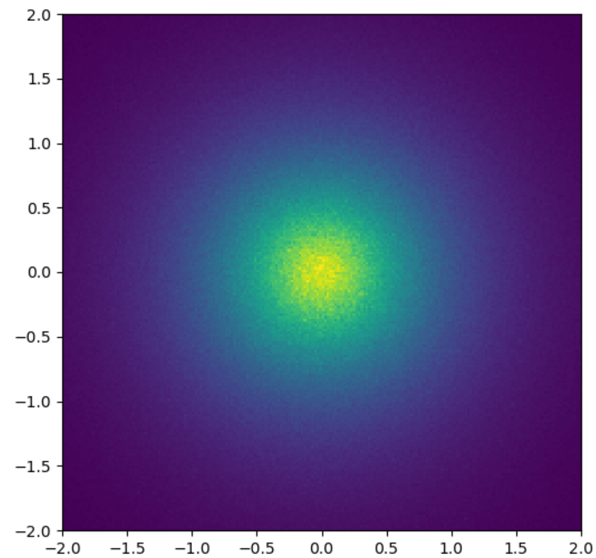

Fig. 3. Histogram of the ground state PDF of the Hydrogen atom. Plotted are the x and y coordinates of samples (projection) taken from the PDF in units of Bohr radii. The points are sampled from 5 Metropolis chains each of length 10^7. There are $200 \times 200 = 4 \times 10^4$ bins in this histogram.

C. 3D Hydrogen Molecule

The Morse Potential for the Hydrogen molecule can be seen in Fig. 4. The values of the constants D, a and r_0 in the Morse potential are 0.137 ± 0.003, 1.24 ± 0.02 and 1.40 ± 0.01, respectively.

The wavefunction of the Hydrogen molecule of the minimum energy state at a certain atomic separation can be visualised in Fig. 5.

The initial value of θ that was chosen was randomly chosen in the range 1 ± 0.05. It can be shown that the minimum is in this range in Fig.6.

IV. Discussion

A. 1D Quantum Harmonic Oscillator

The methods used to obtain the energy eigenstates were Monte Carlo Integration, the Metropolis algorithm and a finite difference method. The potential

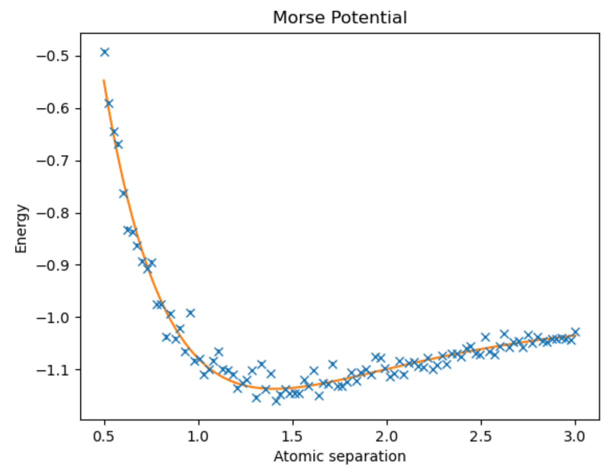

Fig. 4. The Morse Potential. 100 points were taken on the graph from $r = 0.5$ to $r = 3$. In orange is the curve fit of the Morse potential through these 100 points. Each sample was optimised using gradient descent until the step size was below 10^{-3}.

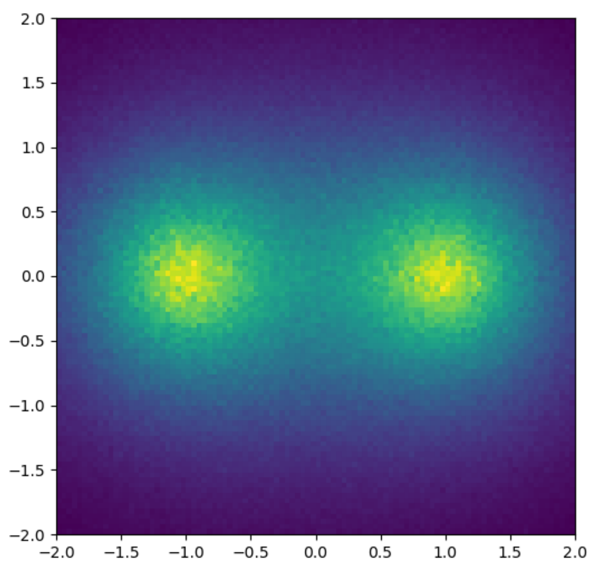

Fig. 5. A histogram of the PDF for the Hydrogen molecule. The nuclei are located at $(\pm 1, 0, 0)$. There are $100 \times 100 = 10^4$ histogram bins and the points were generated from 100 Metropolis chains each of length 2×10^5, and their x and y positions are plotted, projected onto a 2D plane. The wavefunction is the one which minimises energy at this atomic separation.

issues with Metropolis are bias and correlation between data points in the case where the algorithm does not have enough samples to correctly represent the distribution. To counteract this, 5 Metropolis chains were used, each with 10^5 points. The error in the sample mean is smaller than the error in a single

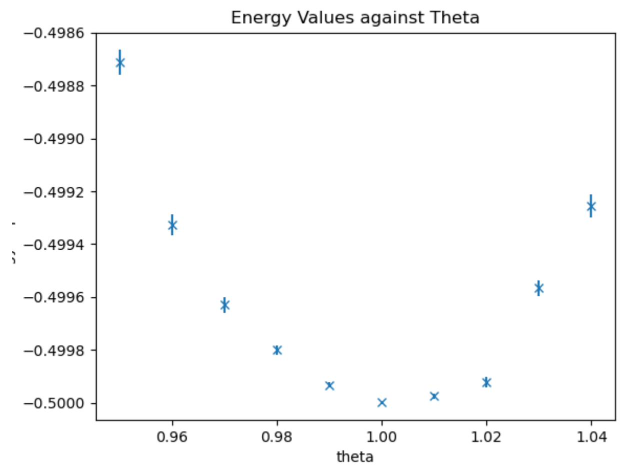

Fig. 6. Plot of energy against θ. Each point on the graph was calculated using 10^5 samples. This justifies the initialisation of the gradient descent.

point by a factor of $10^{-2.5}$. This is true only under the assumptions that the error in a single point is not biased upwards or downwards and that the points are independent. The final source of error is the finite difference method used. For this 1D problem, a second-order finite difference scheme was used to calculate the second derivative with a step size of 10^{-5}.

B. 3D Hydrogen Atom

The potential flaws of the Metropolis algorithm in 1D are the same as those in 3D, but the risk is much higher due to the ”curse of dimensionality”. It takes more points to create a representative sample of the PDF than it takes in 1 dimension. This means that the formula for the error in the sample mean is at risk of giving an underestimation of the true error if not enough points are sampled.

The energy was calculated using a fourth-order finite difference approximation of the Laplacian with a step size of 10^{-6} and the derivative of the energy $\partial_\theta \langle H \rangle$ was calculated analytically.

The gradient descent adaptively sampled more points as the gradient decreased and the threshold condition for the optimisation was

$$(|\partial_\theta \langle H \rangle| + |\Delta \partial_\theta \langle H \rangle|)\alpha < 10^{-6}. \qquad (22)$$

C. 3D Hydrogen Molecule

The gradient descent again adaptively sampled more points as the gradient decreases, but the number of Metropolis chains remained fixed at 25. The threshold condition for this problem was now

$$(|\nabla\langle H\rangle| + |\Delta(\nabla_\theta\langle H\rangle)|)\alpha < 10^{-2}, \tag{23}$$

because this problem takes a lot more compute power and time to converge. The gradient $\nabla\langle H\rangle$ was again calculated analytically, and the same difference scheme was used as for the atom.

V. Conclusion

In this paper, we discussed the process of finding the Morse potential. We found the bond length and the dissociation energy to be 1.40 ± 0.01 and 0.137 ± 0.003. The calculated bond length aligns with the experimental value. However, the calculated dissociation energy does not.

References

[1] M. Scott and J. Owen, *Project Instructions*, Imperial College London, 2025.

[2] A. Buchholz, F. Wenzel, and S. Mandt, "Quasi-Monte Carlo Variational Inference," in *Proceedings of the 35th International Conference on Machine Learning*, J. Dy and A. Krause, eds., vol. 80 of *Proceedings of Machine Learning Research*, pp. 668–677, PMLR, 2018.